Supplementary Materials

From Material Turnover to Causal Memory: Organizational Continuity in a Simulated Droplet

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1 S1. Protocol history (the nine-experiment ladder)

Each experiment was preregistered with a sealed prospective family, executed once, and committed to an isolated version-control branch. HMC (material continuity): snapshot nearly sufficient, no strong history-specific hidden persistence. HSI (hidden state): generic causal attractor, not individuating (AUC 0.3–0.4). IOM-00: designed two-component memory; causal, erasable, transplantable; readout ≈1-D. MCM-00: m_- →attractant makes order readable (∼70/73× ablation collapse). WD-01 Phase B: grouped-evaluation correction (0.57 → 0.19); rank ≈1. Phase C: desaturated writing (C1c); two coordinates at rest, causal ≈1-D. SMC-01: h_2 frame-free dispersion; first (later-corrected) turnover collapse. DMM-01: correction — family variance, not smoothing/dilution; underpowered. H2-CERT-01: sealed kill-switch, $h_2 \approx 0$ at $M \approx 0.21$. TCA-01: tracker-continuity audit; longitudinal re-certification of h_1.

2 S2. Branch / commit registry

experiment	result commit
HMC	25c419a
HSI	709f963
IOM-00	0ea1250
MCM-00	5841b9b
WD-01 Phase B	9a39f8c
WD-01 Phase C (seal / result)	be08738 / a07b95b
SMC-01 (seal / result)	d0f47b6 / 3dc8c35
DMM-01 (seal / result)	5bd46e7 / 49e96ad
H2-CERT-01 (seal / result)	5e1196f / 1f8f789
consolidation / observer	112aace / 6891ef8
tracker audit / post-incident	568af39 / 7afd6e4
short-paper V0 / full manuscript	a97909a / (this commit)

Main branch never modified (HEAD f3921a4); nothing pushed (origin/main e17f431).

3 S3. Sealed prospective family hashes (SHA-256, generated before selection, executed once)

experiment	seeds (dev / prospective)	SHA-256
Phase C	34001–3 / 35001–3	c6d0cd3c5b82aa122ac7e5649888fbc53ca6faa8db68e2c1b48863e87fb9e202
SMC-01	36001–3 / 36501–3	2d6986fe2499cb52fdc309f5b1867cff56cb04b4bd7884633c4c2a148d5792ef
DMM-01	37001–3 / 37501–3	923d8890ea25ab5a136078486f282ad8e07f4cdbaa125da40ef930f5bb8a00b6
H2-CERT	— / 38501–4	4265b98cacb705a7bea56a2e0a5bbe36e10fec3d977e7da6e655037189193fee

Seed families disjoint across experiments (32xxx–38xxx).

4 S4. Parameter grids and negative candidates

Writing-redesign candidates compared at development (Phase C): C0 (frozen baseline), C1a/C1b/C1c (dynamic-range corrections varying $\eta_w, \eta_{d,1}, \eta_{d,2}, k_{\text{exp}}$), C2a (distinct write signals). Selected: C1c, on $\min(R^2(h_1), R^2(h_2))$ subject to viability. Maintenance candidates (DMM Phase A, diagnostic only): templating and diffusion ablations, static memory relaxation; all left the h_2 decode essentially unchanged. Negative variants were retained, not discarded.

5 S5. Gate certificates (machine-readable)

JSON gate certificates are committed per experiment: `results/wd01/wd01_gate_certificate.json`, `results/wd01_phasec/wd01_phasec_gate_certificate.json`, `results/smc01/smc01_gate_certificate.json`, `results/dmm01/dmm01_gate_certificate.json`, `results/h2cert/h2cert_gate_certificate.json`.

6 S6. Tracker unit tests (TCA-01)

(i) Selection audit: the historical rule is arg max size, reselected per frame; five hard switches on seed 38501. (ii) Periodic translation invariance: whole-world shifts (including boundary-straddling) leave size and memory statistics bit-invariant. (iii) Sensitivity: h_1 decodes 0.96/0.98/0.99 from largest / second-largest / longitudinal. (iv) Hold-out: untouched seeds 38502–38504, 36/36 survival, h_1 deep $R^2 = 0.98$ CI $[0.97, 1.00]$, $h_2 = 0.34$. Specification: `docs/audit/TCA_01_TRACKER_SPEC.md`.

7 S7. Seed-family results (h1 turnover)

DMM pool (4 seeds): h_1 $0.98 \to 0.99 \to 0.99$ over $M = 1.00 \to 0.58 \to 0.36$. H2-CERT sealed (4 seeds): $0.94 \to 0.93 \to 0.93$ over $M = 1.00 \to 0.44 \to 0.21$. Longitudinal hold-out (untouched seeds): deep 0.98 CI $[0.97, 1.00]$.

8 S8. Reproduction commands

Environment: python3, numpy, scipy, matplotlib; `PYTHONPATH=$REPO:$REPO/results/wd01_phasec`. Per experiment, resumable runners are committed beside their data: `results/wd01/run_diag.py`; `results/wd01_phasec/{dev_runner,prosp_runner,causal_runner,causal_inplace}.py`; `results/smc01/{fie` `results/dmm01/{phaseA,pooled,prosp}.py`; `results/h2cert/{pilot,cert}.py`; tracker hold-out reruns deterministically from the H2-CERT manifest seeds 38502–38504. Viewer: `scripts/observe_droplet.p` `--preset story`.

9 S9. Raw-data and figure registry

Raw: `results/{sc_mcm,wd01,wd01_phasec,smc01,dmm01,h2cert,observer}/*.pkl,*.json`. Figures (paper): Fig. 1 fig1_model_ladder; Fig. 2 fig2_h1_causal; Fig. 3 tracked_3panel; Fig. 3b fig3b_longitudinal_certific Fig. 4 fig4_order_m_minus; Fig. 5 phasec_figure; Fig. 6 h2cert_figure; Fig. 7 paper_claim_ladder. Each traces to a committed generator script.